

SOC612: Data analysis and visualization (I): Descriptive and exploratory analysis [Task 2]

September 22, 2026

1 Set Up

- **Intro:** The following task is based on the third SOC612 lecture. Please write your solutions either in a Python notebook file (*.ipynb*) or in an R markdown file (*.Rmd*). A script with a possible solution will be available on the Lingnan University moodle, or on GitHub.
- **Topic:** Eating and drinking behaviours of a US sample
- **Data:** National Health Interview Survey-Wave 2022
- **Use of AI:** You may use AI. If so, please indicate how it is used specifically.

2 Descriptive and exploratory analysis

1. Set up a working directory and load the dataset *nhis-22.csv*. This is a prepared, clean dataset, you don't need to do any data cleaning at this point.
2. We try to find out more about the eating habits in our sample. For the variables *fries*, *potato*, *beans*, and *pizza*, calculate the following values, round by 2 digits, and put them into a table called *eating_habits*: mean, median, sd, and range.
3. For the variables *fruit*, *salad*, and *veggie*, create the following plots:
 - fruit: a histogram, color is *light green*. Choose a reasonable number of bins.
 - veggie: a density plot, separated by *gender*. Choose the colours yourself.
 - salad: a boxplot, separated by *education*.

Describe each of the graphs in one, two sentences.

4. We are also interested in the consumption of coffee and soda. However, as we can see above, numeric variables might not be the most illustrative.

- In a first step, turn the two numeric variables *coffee* and *soda* into factors called *coffee.cat* and *soda.cat*. Use the following transformation: 0="never"; 1-3="rarely"; 4-10="sometimes"; 11-20="regularly"; 21-29="often", 30="daily", 30+="daily+".
- Create a bar chart for *soda.cat* by gender.
- Create a bar chart for *coffee.cat* by education.

Describe each of the graphs with one, two sentences.

- Now we're interested in the correlation between all eating and drinking variables.
 - First, calculate the correlations of all food- and drinking-related behaviours.
 - Second, visualize the correlations.

Describe the correlation plot in one, two sentences.

- Finally, we assume that there might be some outliers in the variables *pizza*, *salsa*, and *tomato_sauce*. Use one of the standard procedures for outliers detections. How many cases are outliers on all of these three together?

Good luck!